

Proposition: Selecting any n integers from the set $\{1, 2, 3, \dots, 20\}$ will result in at least one of the n dividing another.

Observe that any integer can be represented as the product of some odd integer and a power of 2. The set

1, 2, 3, 4, 5, 6, 7, 8, 9, 10
11, 12, 13, 14, 15, 16, 17, 18, 19, 20

becomes

$1 \cdot 2^0, 1 \cdot 2^1, 3 \cdot 2^0, 1 \cdot 2^2, 5 \cdot 2^0, 3 \cdot 2^1, 7 \cdot 2^0, 1 \cdot 2^3, 9 \cdot 2^0, 5 \cdot 2^1$
 $11 \cdot 2^0, 3 \cdot 2^2, 13 \cdot 2^0, 7 \cdot 2^1, 15 \cdot 2^0, 1 \cdot 2^4, 17 \cdot 2^0, 9 \cdot 2^1, 19 \cdot 2^0, 5 \cdot 2^2$

Each integer is now in the form $\phi \cdot 2^k$ where $\phi \in \mathbb{Z}^{\text{odd}}, k \in \mathbb{Z}$.

Observe that $0 \leq k \leq 4$, and ϕ has one of 10 values: $\{1, 3, 5, 7, 9, 11, 13, 15, 17, 19\}$, which are the ten odd integers from the original set.

Since every number in the original set can be written as $n = \phi \cdot 2^k$, and ϕ can have 10 possible values, at least one ϕ value will appear twice in the 11-integer subset.

The occurrence of two integers sharing a ϕ value indicates that one is divisible by the other.

Suppose

$$\begin{aligned} n_1 &= \phi \cdot 2^{k_1} \\ n_2 &= \phi \cdot 2^{k_2} \end{aligned}$$

Then either $k_1 < k_2$ or $k_2 < k_1$. If $k_1 = k_2$, then $n_1 = n_2$, and this cannot happen since every integer in the original set is unique and no integer may be used twice in forming the 11-value subset.

$$\text{Then } \frac{n_m}{n_i} = \frac{\phi \cdot 2^{k_m}}{\phi \cdot 2^{k_i}} = \frac{2^{k_{\max}}}{2^{k_{\min}}}$$

where $k_{\max}$ = the larger of k_1 and k_2 , and
 $k_{\min}$ = the lesser of k_1 and k_2 .

Since $2^{k_{\max}}$ and $2^{k_{\min}}$ are powers of 2 and
 $k_{\min} < k_{\max}$, $2^{k_{\min}}$ divides $2^{k_{\max}}$.

∴ Choosing any 11 integers from the set of
 integers from 1 to 20 guarantees that at least
 one of these will divide another.

$$\phi = 1 \cdot 2^0$$

$$\phi = 3 \cdot 2^0$$

$$\phi = 5 \cdot 2^0$$

$$\phi = 7 \cdot 2^0$$

$$\phi = 9 \cdot 2^0$$

$$\phi = 11 \cdot 2^0$$

$$\phi = 13 \cdot 2^0$$

$$\phi = 15 \cdot 2^0$$

$$\phi = 17 \cdot 2^0$$

$$\phi = 19 \cdot 2^0$$

11 integers into 10 boxes $\rightarrow$ at least 1 box has 2 or
 more placements. Q.E.D.